

Loan Approval Prediction Case Study

Data Science Internship Project

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1. Introduction

This project aims to predict loan approval status using borrower information. Machine learning models are trained and evaluated to identify applicants likely to receive loan approval while minimizing business risk.

2. Dataset Description

The dataset contains applicant information such as gender, marital status, education, income, loan amount, credit history, and loan approval status. The target variable is Loan_Status.

3. Data Preprocessing

- Missing values handled using median and mode imputation.
- Categorical variables encoded using Label Encoding and One-Hot Encoding.
- Numerical features scaled using StandardScaler.
- Duplicate records removed where applicable.

4. Class Imbalance Handling

The dataset was examined for class imbalance. SMOTE oversampling and class-weight adjustments were applied to improve model performance on minority classes.

5. Models Implemented

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

6. Evaluation Metrics

Models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Confusion matrices and ROC curves were also analyzed.

7. Key Findings

- Credit history is one of the strongest predictors of loan approval.
- Applicant income positively influences approval probability.
- Handling class imbalance improves recall performance.
- Random Forest generally provides superior predictive performance.
- Logistic Regression offers better interpretability.

8. Business Interpretation

High recall ensures that eligible customers are not unnecessarily rejected, while high precision reduces the risk of approving high-risk applicants. Business objectives determine the appropriate balance between these metrics.

9. Recommended Deployment Strategy

Recommended Model: Random Forest Classifier

Suggested Probability Threshold: 0.60

Reason: Provides a balanced trade-off between recall and precision while reducing the risk associated with false approvals.

10. Conclusion

The developed machine learning pipeline demonstrates the feasibility of automating loan approval decisions. Proper preprocessing, imbalance handling, and threshold tuning significantly improve model effectiveness.